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## Smart Shopping Cart

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## TARGETS OF THE MINI PROJECT WORK

| ACTIVITIES                            |     | ACTIVITIES             |     |
|---------------------------------------|-----|------------------------|-----|
| Domain & problem identified           | Yes | Development of product | Yes |
| Literature Review                     | Yes | Testing                | Yes |
| Objectives formulated                 | Yes | Obtained Result        | Yes |
| Methodology / Design                  | Yes | Documentation          | Yes |
| Created work plan and task allocation | Yes | Report submission      | Yes |



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## INTRODUCTION

- Smart Shopping Cart aims to enhance the shopping experience by reducing time spent at billing counters.
- Traditional billing relies on barcode scanning, requiring manual effort and increasing checkout time.
- The proposed system uses RFID tags instead of barcodes to streamline the shopping process.



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## INTRODUCTION

- Uses an RFID reader with an LCD display mounted on the shopping cart to display the price of each scanned item.
- Improves overall customer satisfaction and operational efficiency in shopping malls.



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## OBJECTIVES

- Develop an RFID-based smart shopping cart to automate item detection and displaying item details and total cost in real time.

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## EXISTING SYSTEM

- The existing system relies on barcode scanning, requiring each item to be scanned manually at the checkout counter.
- Manual scanning in traditional billing systems increases the chances of human errors, such as scanning the wrong item, multiple scans, or missed items, resulting in inaccurate billing.

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## EXISTING SYSTEM

- The dependency on barcode scanners means that damaged or missing barcodes can cause delays and require manual price entry.
- Customers need to remove items from their carts for scanning, which adds inconvenience and increases the overall shopping time.



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## LITERATURE REVIEW

| SL NO | Name of the Paper                                                                                                       | Outcome                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Author and Publication Details                                                                                                          |
|-------|-------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| 1     | Data transmission using RFID system on smart shopping carts for checkout process efficiency in supermarket at indonesia | <ul style="list-style-type: none"><li>The project resulted in the successful implementation of a Smart Shopping System, which reduced queue times by allowing customers to self-scan items, thus making the checkout process more efficient and convenient.</li><li>Customers need to familiarize themselves with the self-scanning feature and smartphone app, including scanning items, checking prices, and managing shopping lists. Some may face a learning curve while adapting to this technology.</li></ul> | Martinusa , Metta Saridewi Wahaba , Yudia, Hanry Ham, published in the journal Procedia Computer Science, volume 179 on January 1, 2021 |



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| SL NO | Name of the Paper                         | Outcome                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Author and Publication Details                                                                                                                                                           |
|-------|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2     | Smart shopping cart using RFID technology | <ul style="list-style-type: none"><li>In this system, the customer begins by logging in via a QR code on the shopping cart, enabling a seamless shopping experience. The cart automatically detects items using RFID technology, updates the bill in real-time, and manages stock efficiently.</li><li>A challenge in this system is ensuring user-friendly QR code login process, as some customers may face difficulties with scanning, app compatibility, or network issues, potentially delaying their shopping experience.</li></ul> | Susanna M Santhosh, Cemone P Babu, Jills Joseph , Preethu Benny, published in the International Journal of Engineering Applied Sciences and Technology (IJEAST), Volume 5, on June 2020. |

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| SL NO | Name of the Paper                        | Outcome                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Author and Publication Details                                                                                                                                                     |
|-------|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3     | IOT based Smart Shopping Cart using RFID | <ul style="list-style-type: none"><li>• This system utilizes RFID technology, an Arduino microcontroller, and a Bluetooth module to track items in real time and automate billing. It reduces checkout times, provides product details, and updates inventory by recording purchases and generating bills automatically.</li><li>• The main challenges faced in this project include high initial setup costs, customer adaptation to new technology, ensuring data privacy and security, and the need for regular maintenance and updates.</li></ul> | Seshagopal , Harish D , Ashwinth Fatin , Dr. Antonidoss , published in the International Journal of Engineering Research & Technology (IJERT), Volume 10, Issue 04, in April 2021. |

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| SL NO | Name of the Paper                                                   | Outcome                                                                                                                                                                                                                                                                                                                                                                                                                                        | Author and Publication Details                                                                                                                                                                      |
|-------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4     | RFID Based Smart Shopping Cart for Smart Shopping Enhanced with IoT | <ul style="list-style-type: none"><li>This RFID-based system automates billing by scanning tagged products and displaying details on an LCD screen. It reduces wait times and uses GSM modules for real-time updates and notifications.</li><li>A challenge in this system is maintaining stable GSM connectivity, as network issues or poor signal strength may disrupt real-time, affecting the efficiency of the checkout process</li></ul> | K Sreenivasa Rao, Ayesha Shaik, N Dineshsairamreddy, H Anusha, A Guru Rajesh, published in the Journal of Emerging Technologies and Innovative Research (JETIR), Volume 10, Issue 3, in March 2023. |

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| SL NO | Name of the Paper               | Outcome                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Author and Publication Details                                                                                                                                |
|-------|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5     | Research on Smart Shopping Cart | <ul style="list-style-type: none"><li>• This project develops an IoT-based Smart Shopping Cart using RFID technology to automate billing and enhance the shopping experience. It features automatic item detection, reduces checkout wait times. The system integrates an microcontroller, GSM module, LCD, and keypad to display costs.</li><li>• challenge for this system is ensuring seamless communication between the GSM module and the server, as network disruptions or delays could affect real-time transaction updates and payment notifications, leading to a less efficient shopping experience.</li></ul> | Prof. Roopa C , Nivas Chandra Reddy,<br>published in the Journal of Emerging Technologies and Innovative Research (JETIR), Volume 11, Issue 3, in March 2024. |

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## PROPOSED SYSTEM

- RFID Scanning: Scan items as you add them to the cart for instant recording.
- Real-Time Display: LCD screen displays item details and updated totals.
- Automated Billing: Cart maintains a live total for fast and easy checkout.
- No Manual Scanning: Eliminates manual barcode scanning, reducing cashier workload and wait times.



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## ADVANTAGES

- Reduced Checkout Stress
- Efficient Multi-Item Scanning
- Increased Customer Satisfaction



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# METHODOLOGY

## **Module 1: RFID-Based Product Identification Module**

Objective: Automatically detect and identify products added to the cart

Components: RFID Tags → RFID Reader → Microcontroller

## **Module 2. Billing & Pricing Module**

Objective: Maintain an updated total bill as products are added/removed.

Components: Microcontroller → Display Unit



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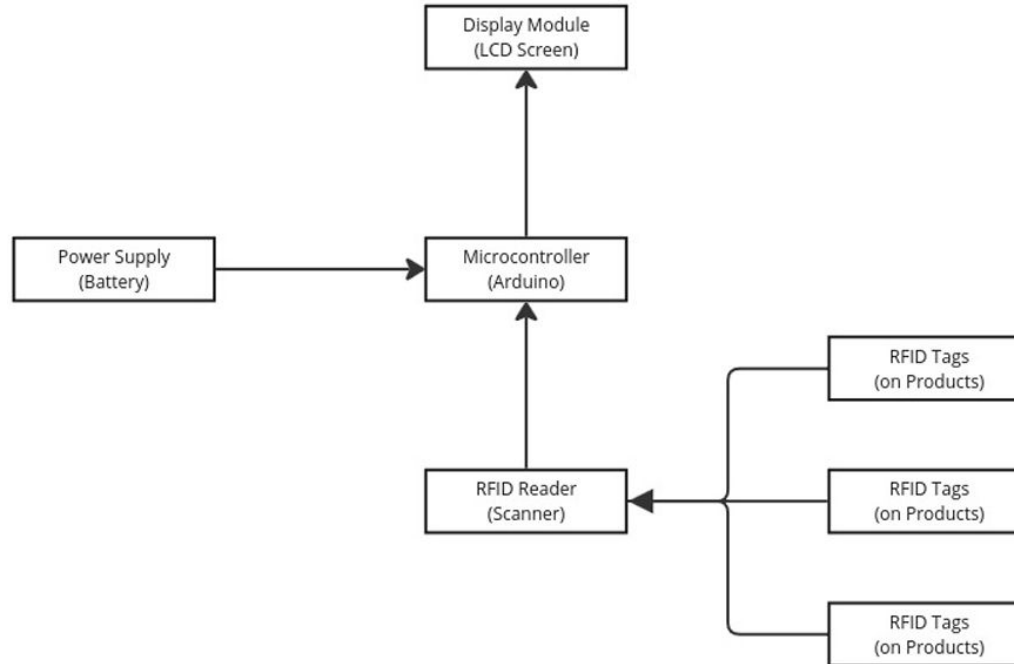
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## BLOCK DIAGRAM



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## LEVEL 0 BLOCK DIAGRAM



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## LEVEL 1 BLOCK DIAGRAM



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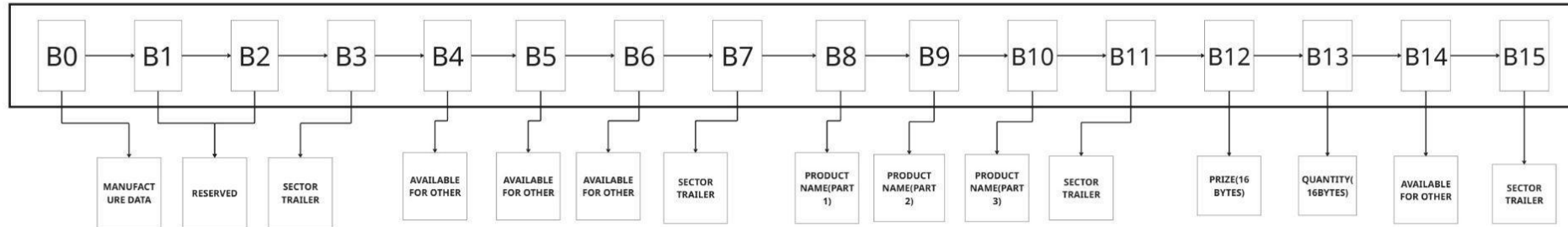
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## LEVEL 2 BLOCK DIAGRAM



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## RESOURCES NEEDED

### SOFTWARE REQUIREMENTS:

- Arduino IDE

### HARDWARE REQUIREMENTS:

- RFID Reader (Mounted on the cart)
- RFID Tags (Attached to products)
- Microcontroller (Arduino)
- Display Unit (LCD)
- Power Supply (Rechargeable battery)

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## WORK PLAN

|                |                                    |
|----------------|------------------------------------|
| ABIN SANTHOSH  | Development & Implementation       |
| ADHITH SUNIL   | Project Monitoring & Quality Check |
| ADHWAITH T T   | Testing & Risk Management          |
| ANITTA RAPHI E | Documentation & Resource Handling  |





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## FEASIBILITY ANALYSIS

- Technical Feasibility
- Economic Feasibility
- Social Feasibility
- Legal & Ethical Feasibility

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# CODING SNAPS

```
new-working-full-2 | Arduino IDE 2.3.4
File Edit Sketch Tools Help

new-working-full-2.ino
1 #include <SPI.h>
2 #include <MFRC522.h>
3 #include <U8g2lib.h>
4
5 #define RST_PIN 9
6 #define SS_PIN 10
7
8 U8G2_SH1106_128X64_NONAME_F_HM_I2C u8g2(U8G2_R0);
9
10 MFRC522 rfid(SS_PIN, RST_PIN);
11 MFRC522::MIFARE_Key key;
12
13 const byte MAX_ITEMS = 5; // Increased capacity since we're not storing names
14 struct InventoryItem {
15     byte uid[4]; // 4 bytes for UID
16     float price; // 4 bytes for price
17 };
18 InventoryItem inventory[MAX_ITEMS];
19 byte itemCount = 0;
20 float totalValue = 0.0;
21
22 void setup() {
23     Serial.begin(9600);
24     SPI.begin();
25     rfid.PCD_Init();
26
27     u8g2.begin();
28     u8g2.setFont(u8g2_font_6x10_tr);
29
30     for (byte i = 0; i < 6; i++) {
31         key.keyByte[i] = 0xFF;
32     }
33
34     showDisplayMessage("System Ready");
```

```
new-working-full-2 | Arduino IDE 2.3.4
File Edit Sketch Tools Help

new-working-full-2.ino
35 }
36
37 void loop() {
38     if (!rfid.PICC_IsNewCardPresent() || !rfid.PICC_ReadCardSerial()) {
39         delay(50);
40         return;
41     }
42
43     byte uid[4];
44     memcpy(uid, rfid.uid.uidByte, 4);
45     byte itemIndex = findItemIndex(uid);
46
47     // Read product name from tag (but don't store it)
48     char productName[12];
49     bool hasName = readProductName(productName);
50     float price = readProductPrice();
51
52     if (itemIndex == 0xFF && itemCount < MAX_ITEMS && price > 0) {
53         addItem(uid, price);
54         showLiveTagInfo("ADDED", productName, price, hasName);
55     }
56     else if (itemIndex != 0xFF) {
57         float removedPrice = inventory[itemIndex].price;
58         removeItem(itemIndex);
59         showLiveTagInfo("REMOVED", productName, removedPrice, hasName);
60     }
61
62     rfid.PICC_HaltA();
63     rfid.PCD_StopCrypto1();
64     delay(1000);
65 }
66
67 // Display live tag info (name is read fresh each time)
68 void showLiveTagInfo(const char* action, const char* productName, float price,
```

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# CODING SNAPS

```
new-working-full-2 | Arduino IDE 2.3.4
File Edit Sketch Tools Help

new-working-full-2.ino

136 byte pos = 0;
137 memset(buffer, 0, 12);
138
139 for (byte block = 8; block <= 10 && pos < 11; block++) {
140     byte sector = block / 4 * 4 + 3;
141     if (rfid.PCD_Authenticate(MFRC522::PICC_CMD_MF_AUTH_KEY_A, sector, &key, &{
142         if (rfid.MIFARE_Read(block, sectorBuffer, &size) == MFRC522::STATUS_OK) {
143             for (byte i = 0; i < 16 && pos < 11; i++) {
144                 if (sectorBuffer[i] > 31 && sectorBuffer[i] < 127) {
145                     buffer[pos++] = (char)sectorBuffer[i];
146                 }
147             }
148         }
149     }
150 }
151 return (pos > 0); // Returns true if name was read
152 }
153
154 float readProductPrice() {
155     byte buffer[18];
156     byte size = sizeof(buffer);
157     char priceStr[8];
158     byte pos = 0;
159
160     if (authenticate(12)) {
161         if (rfid.MIFARE_Read(12, buffer, &size) == MFRC522::STATUS_OK) {
162             for (byte i = 0; i < 16 && pos < 7; i++) {
163                 if (buffer[i] >= '0' && buffer[i] <= '9' || buffer[i] == '.') {
164                     priceStr[pos++] = (char)buffer[i];
165                 }
166             }
167             priceStr[pos] = '\0';
168             return atof(priceStr);
169         }
170     }
171 }
```

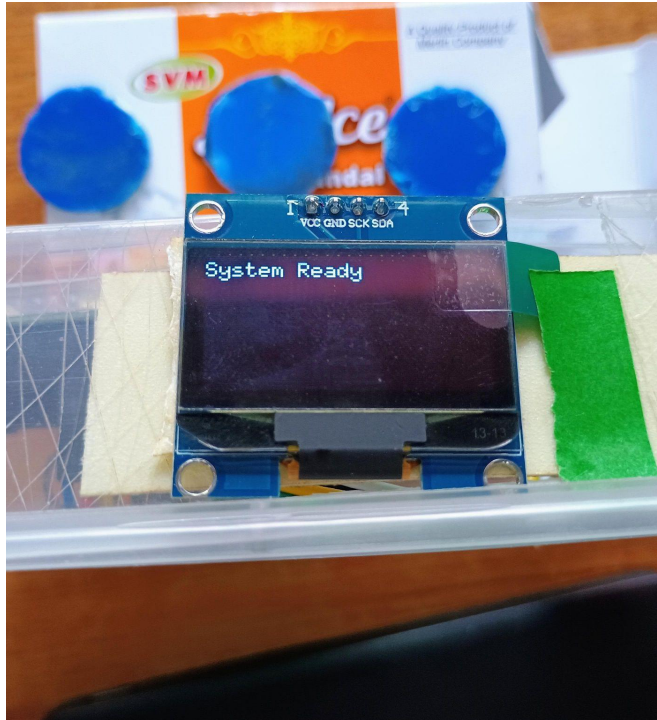
```
new-working-full-2 | Arduino IDE 2.3.4
File Edit Sketch Tools Help

new-working-full-2.ino

150 }
151 return (pos > 0); // Returns true if name was read
152 }
153
154 float readProductPrice() {
155     byte buffer[18];
156     byte size = sizeof(buffer);
157     char priceStr[8];
158     byte pos = 0;
159
160     if (authenticate(12)) {
161         if (rfid.MIFARE_Read(12, buffer, &size) == MFRC522::STATUS_OK) {
162             for (byte i = 0; i < 16 && pos < 7; i++) {
163                 if (buffer[i] >= '0' && buffer[i] <= '9' || buffer[i] == '.') {
164                     priceStr[pos++] = (char)buffer[i];
165                 }
166             }
167             priceStr[pos] = '\0';
168             return atof(priceStr);
169         }
170     }
171     return 0.0;
172 }
173
174 bool authenticate(byte block) {
175     byte sector = block / 4 * 4 + 3;
176     return rfid.PCD_Authenticate(MFRC522::PICC_CMD_MF_AUTH_KEY_A, sector, &key, 8
177 }
178
179 void showMessage(const char* message) {
180     u8g2.clearBuffer();
181     u8g2.drawStr(0, 10, message);
182     u8g2.sendBuffer();
183 }
```



## OUTPUT







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# OUTPUT



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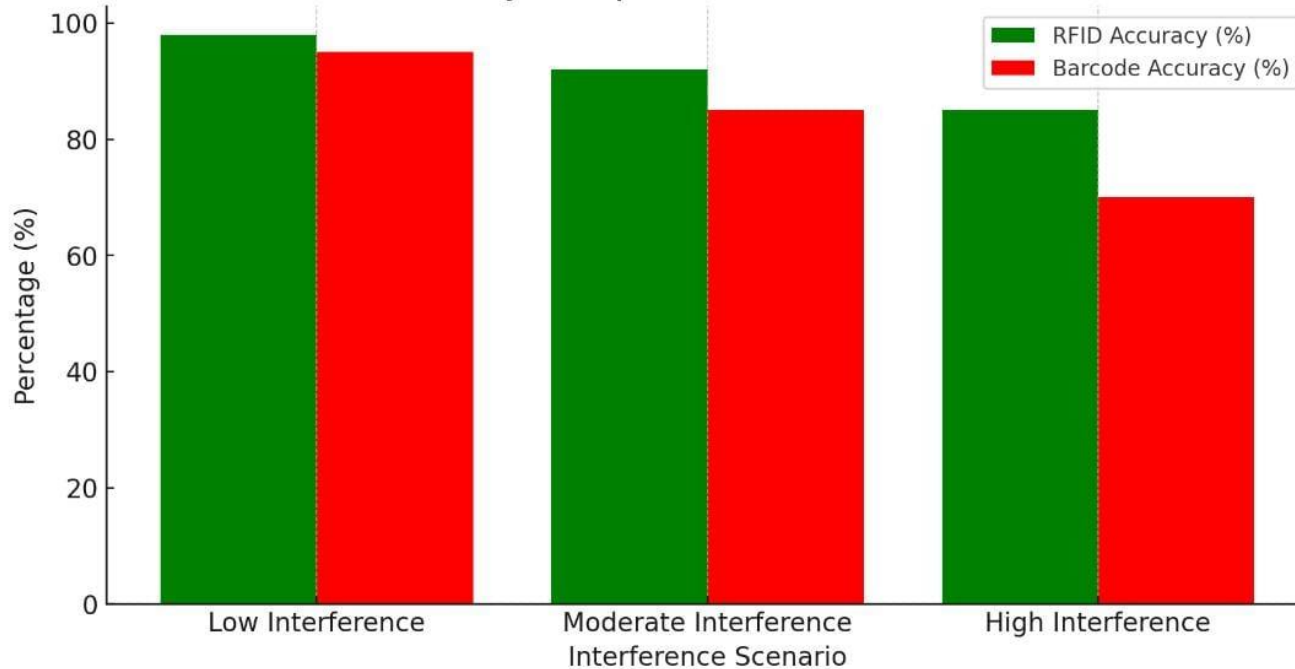
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# PERFORMANCE ANALYSIS

Accuracy Comparison: RFID vs Barcode



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## TESTING METHODS

- Unit Testing (Individual Components): Test RFID detection, data writing/reading, display messages, and price calculations.
- Integration Testing (Component Interaction): Verify RFID and Arduino communication, OLED display updates, and duplicate scan handling.
- Functional Testing (End-to-End System Check): Ensure correct product addition/removal, cart total updates, and error handling.



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## TESTING METHODS

- Stress Testing (Performance Under Load): Test multiple tag scans, power failure recovery, and system response time.
- User Experience Testing (Ease of Use): Check read distance, angles, and overall user-friendliness.
- Security Testing (Prevent Unauthorized Access): Test with unauthorized tags and detect data tampering attempts.

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## CONCLUSION

- We planned our work before implementation.
- The problem statement was defined, and the methodology was finalized.
- All necessary components were procured.
- The Smart Shopping Cart using RFID technology was successfully implemented.
- It improves shopping convenience and retail efficiency.
- Despite some challenges, it offers benefits like faster checkout and better inventory management.

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## REFERENCES

- [1] Susanna M Santhosh, Cemone P Babu, Jills Joseph , Preethu Benny, published in the International Journal of Engineering Applied Sciences and Technology (IJEAST), Volume 5, on June 2020.
- [2] Martinusa , Metta Saridewi Wahaba , Yudia, Hanry Ham, published in the journal Procedia Computer Science, volume 179 on January 1, 2021.
- [3] Seshagopal , Harish D , Ashwinth Fatin , Dr. Antonidoss , published in the International Journal of Engineering Research & Technology (IJERT), Volume 10, Issue 04, in April 2021.

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## REFERENCES

[4] K Sreenivasa Rao, Ayesha Shaik, N Dineshsairamreddy, H Anusha, A Guru Rajesh, published in the Journal of Emerging Technologies and Innovative Research (JETIR), Volume 10, Issue 3, in March 2023.

[5] Prof. Roopa C , Nivas Chandra Reddy, published in the Journal of Emerging Technologies and Innovative Research (JETIR), Volume 11, Issue 3, in March 2024.

**THANK YOU**